

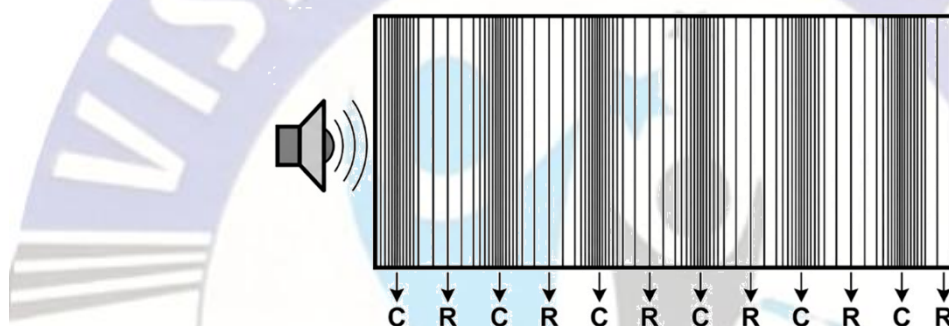
SOUND

• Introduction

- Sound is the one of the sense out of five sense such that hearing, sight, smell, taste and touch.
- When we hearing “Something” reaches our ear is known as “Sound”.
- Sound is a type of energy which makes us to hear.
- Sound is produced when something vibrates.
- It is produced when we speak, strike. The membrane of tabla speaker of a radio
- It is a longitudinal wave.
- It is a mechanical wave i.e, it need a medium to travel.
- It travels in the form of wave.
- It can not travel in the vacuum.

▪ Propagation of Sound

- Sound is propagated by vibration of particles which disturb in the medium.
- Vibration of particles in longitudinal wave.



R - Rarefaction —→ Low density
 → Low pressure

C – Compression —→ High density
 → High pressure

• Wave

- A wave is vibrating disturbance in a medium which travels with energy from one point of space to another point.
- The motion of the disturbance is called the wave motion.

Types of wave on the basis of medium of propagation

1. Mechanical wave.
2. Non – mechanical wave.

1. Mechanical wave

- Those wave which need a medium (S, L, G) to propagate are called mechanical waves.

Example:-

- ✓ Sound waves.
- It does not travel in vacuum.

2. Non - mechanical waves

- Those wave which do not need a material medium (Solid, Liquid, Gas) to propagate are called Non – mechanical waves

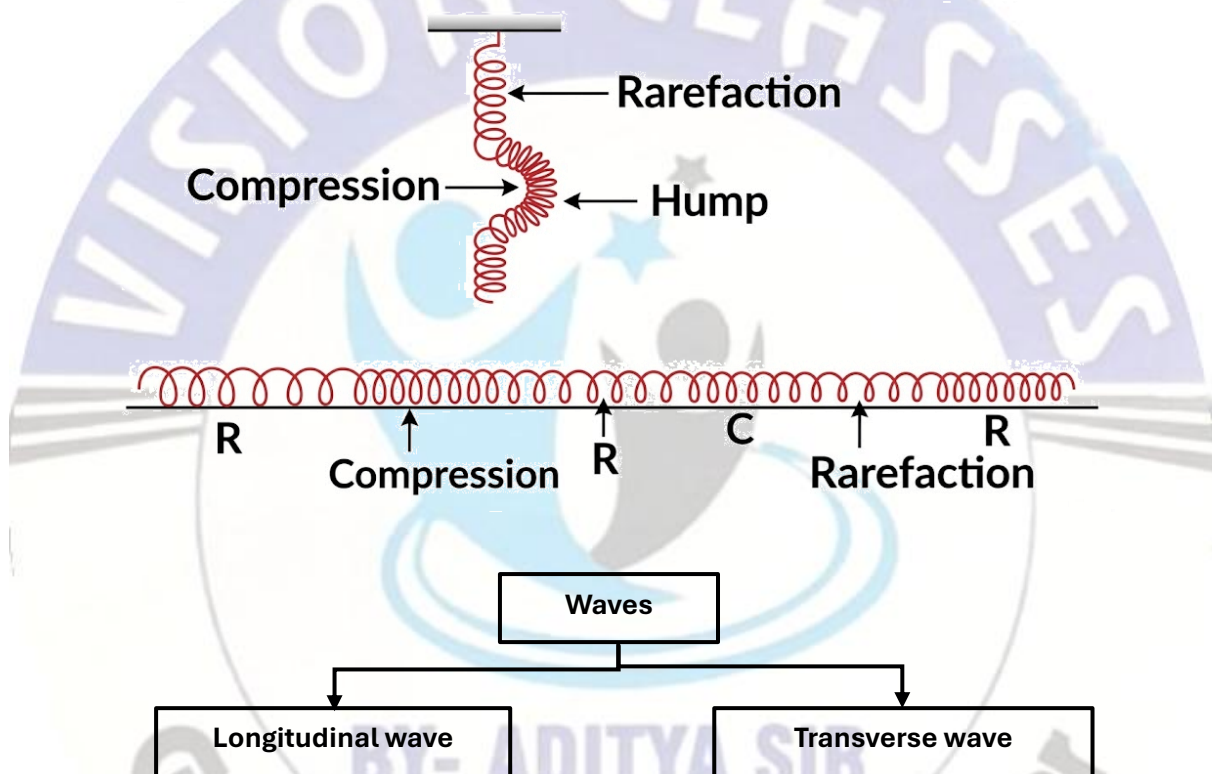
Example:-

- ✓ Light wave, radio wave, T.V waves etc.

- It can travel in the vacuum.

- Spring OR Slinky

- A slinky is a spring shaped toy which can be extended/compressed very easily.
- It is very flexible and can be put into many shapes easily.

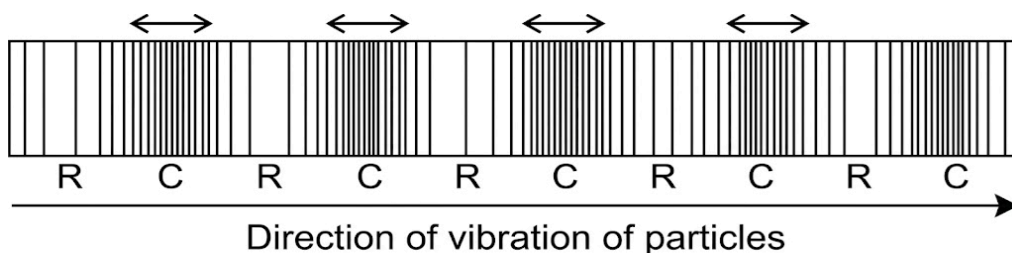


- Longitudinal wave

- That wave in which the particles of a medium move parallel (along) the direction of motion of the propagation of wave, called longitudinal wave.
- The particles of the medium vibrate forth and back in the same direction of the propagation of wave.
- It is a type of mechanical wave.
- It can be produced at all mediums such as Solids, Liquids and Gases.
- The longitudinal wave involves change in the density of the medium.

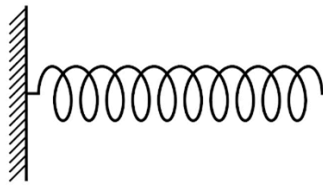
Example:-

- ✓ Ultrasound wave, sound waves, when spring/slinky is pulled/pushed horizontally.

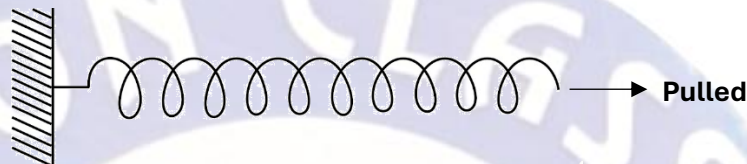


In the case of spring

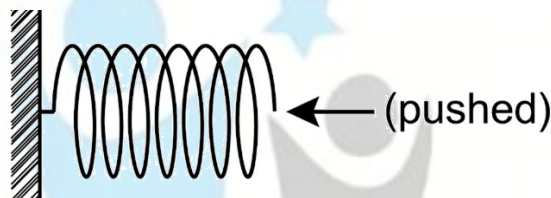
Case 1: Normal shape of the spring



Case 2: Rarefaction of the spring (density decreases)



Case 3: Compression of the spring (i.e, density increases)

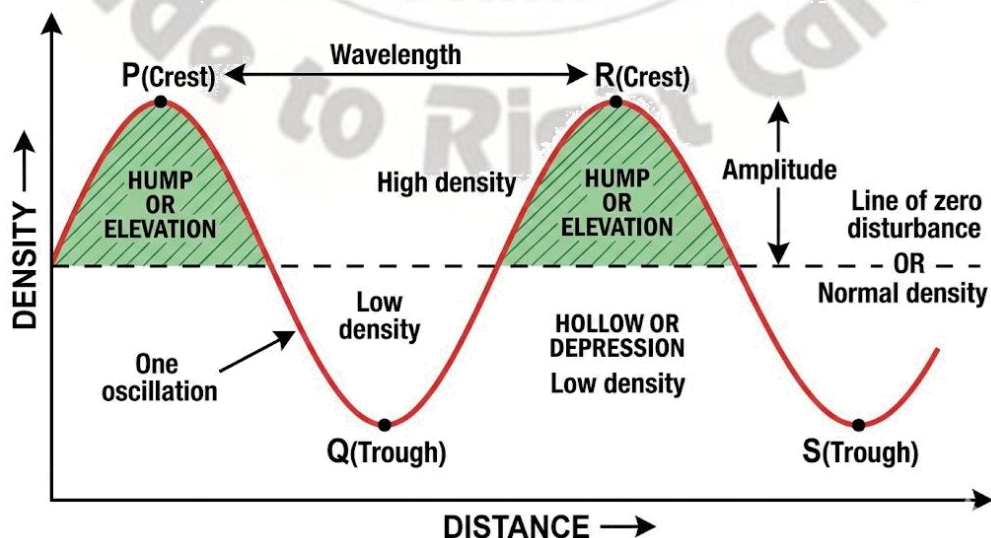


• **Transverse wave**

- That wave in which the particles of a medium move perpendicular to the direction of motion of the propagation of wave, called transverse wave.
- The particles of a medium vibrates up and down (perpendicular) of the direction of the wave.
- It is a non – mechanical wave.
- The transverse waves involve change in the shape of the medium.
- It can be produced in only solids not through liquids or gases and partially on the surface of liquid.

Example:-

- ✓ Light waves, microwaves, Radio waves (Electromagnetic waves), ripples on the surface of water, vibrations in Guitar string etc.



Note:

1. Hump OR Elevation OR greater density.
2. Hollow OR Depression OR less density.

• **Crests**

- The peak or highest point of a wave is called crest.

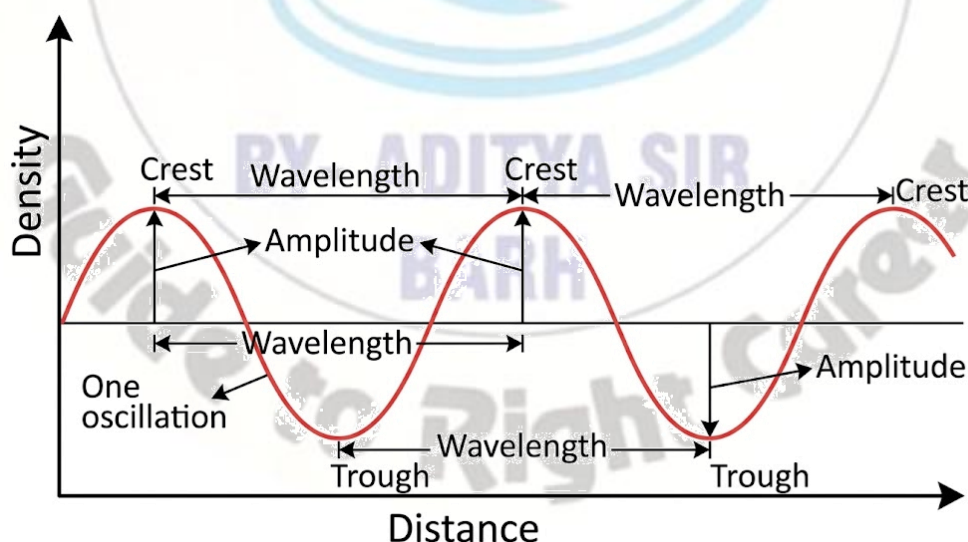
• **Troughs**

- The lowest point of a wave is called Trough.

▪ **Differences between longitudinal wave and transverse wave**

TRANSVERSE WAVE	LONGITUDINAL WAVE
Transverse waves consist of crest and trough.	Longitudinal waves consist of compression and rarefaction.
There are no pressure and vibration.	There is a pressure variation throughout the medium.
They can be propagated through solid and surface of liquid but not in gas.	They can be propagated through solids, liquids as well as gases.
The particles of medium vibrate perpendicular to the direction of wave.	The particles of medium vibrate parallel to the direction of the wave.
There is no change in density of the medium.	There is a change in the density throughout the medium.
Ex – Light wave, T.V wave etc.	Ex – Sound wave, ultrasound wave etc.

▪ **Some important terms**



• **Wavelength**

- The minimum distance after which the state of disturbance is repeated in the wave motion is called wavelength of the wave.

- It is denoted by " λ ".

↳ Lamda (Greek letter)

- The S.I unit of wavelength is meter which is denoted by "m".

- **Time period**

- The time taken to complete one oscillation of the density of the medium is called the time period of the wave.
- It is denoted by "T".
- The S.I unit of time period is "Second" which is denoted by "S".

- **Frequency**

- The number of oscillations of the density of the medium at a place per unit time is called the frequency of the wave.
- It is denoted by "ν" (Nu).

└───────────> Greek letter

$$\therefore \nu = \frac{1}{T} = \frac{1}{\text{Time period}}$$

- The S.I unit of frequency is Hertz which is denoted by "Hz".
 $\therefore 1 \text{ Hz} = 1/\text{s} = 1\text{s}^{-1}$

Note:

- If the no. of complete oscillation of wave in 1 second is 100 then the frequency of the wave is 100 Hz.

- **Speed of the wave/Velocity of the wave**

- The speed with which a disturbance propagates in a medium is called a wave speed.
- The S.I unit of speed of wave is m/s.

Example:-

- ✓ The speed of sound in different mediums.

Medium	Temperature	Speed (m/s)
Dry air	0°C	332 m/s
Dry air	20°C	344 m/s
Distilled water	20°C	1498 m/s
Sea water	0°C	1531 m/s
Copper	20°C	3750 m/s
Aluminum	20°C	5100 m/s
Iron / steel	20°C	5130 m/s
Glass	20°C	5170 m/s
Steel	20°C	5790 m/s

Note:

- The speed of sound depends on three factors.

Case 1: The nature of the material/medium

- The speed of sound is different in different materials.

Example:-

- ✓ Speed of sound: Solid > liquid > gas

Case 2: On the temperature of air

Speed of sound $\propto$ Temperature

Case 3: On the humidity of air

Speed of sound $\propto$ Humidity

- **The relationship between velocity, frequency and wavelength of a wave**

$$\therefore \text{velocity} = \frac{\text{Distance travelled}}{\text{Time taken}}$$

$$\therefore V = \frac{\text{Wavelength}}{T}$$

$$\Rightarrow V = \frac{\lambda}{T}$$

$$\Rightarrow V = \lambda \times \frac{1}{T}$$

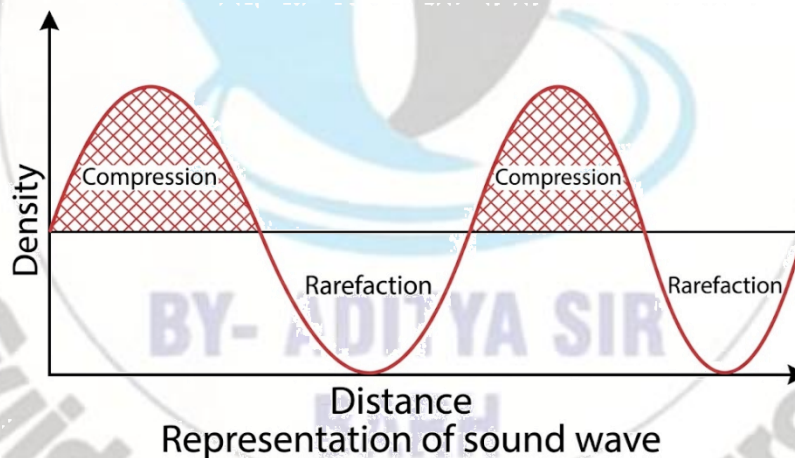
$$\Rightarrow V = \lambda \times \nu \quad \left[\because \frac{1}{T} = \nu \text{ (frequency)} \right]$$

$$\Rightarrow V = \lambda \times f \quad [\nu = f]$$

i.e, velocity of wave = frequency $\times$ wavelength

- **Production of Sound**

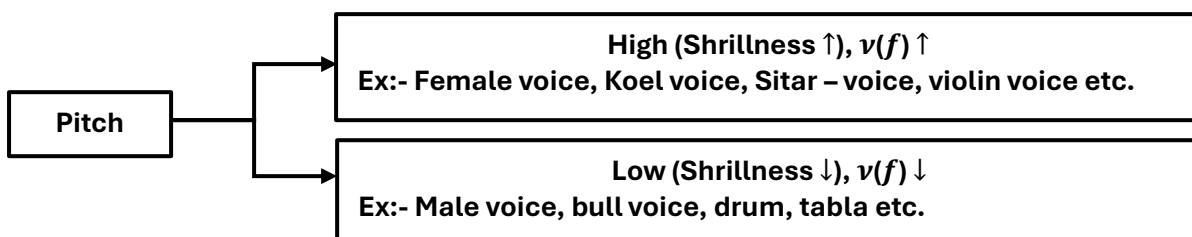
- Sound is produced when an object vibrates. So, sound is produced by vibrating objects, for vibrating energy is required.



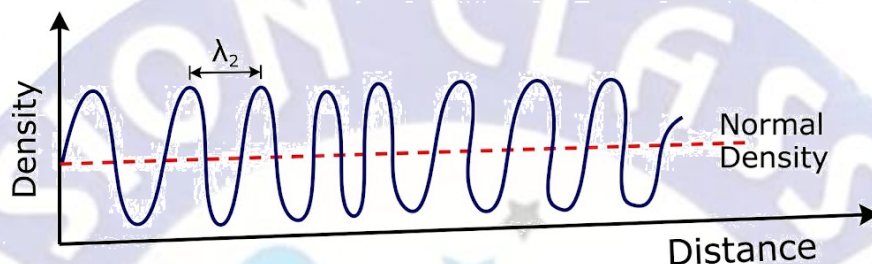
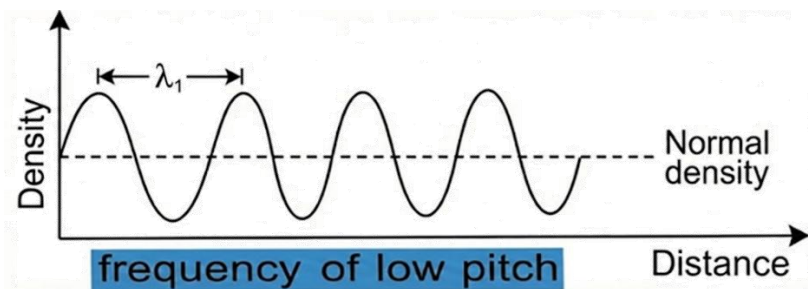
- **Characteristics of sound waves**

1. Pitch

- The quality of a sound governed by the rate of vibrations producing is called pitch.
- Pitch describes how the brain interprets the frequency of sound.
- Pitch is that characteristics of sound by which we can distinguish between different sound of the same loudness.



i.e, Pitch $\propto$ Frequency of Sound



Note:

1. Pitch $\propto \frac{1}{\text{wavelength}}$

Example:- $\lambda_1 > \lambda_2$

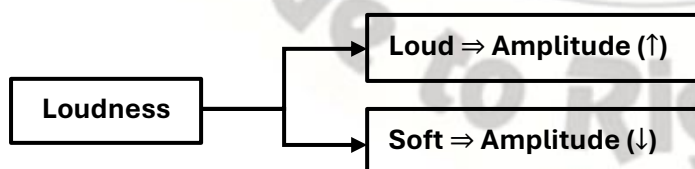
2. The amplitude of the sound wave will remain the same on either increasing or, decreasing the pitch because amplitude of the sound wave is independent of its frequency.

2. Loudness

➤ The loudness of the sound is a phenomenon caused by the amplitude of the sound wave.

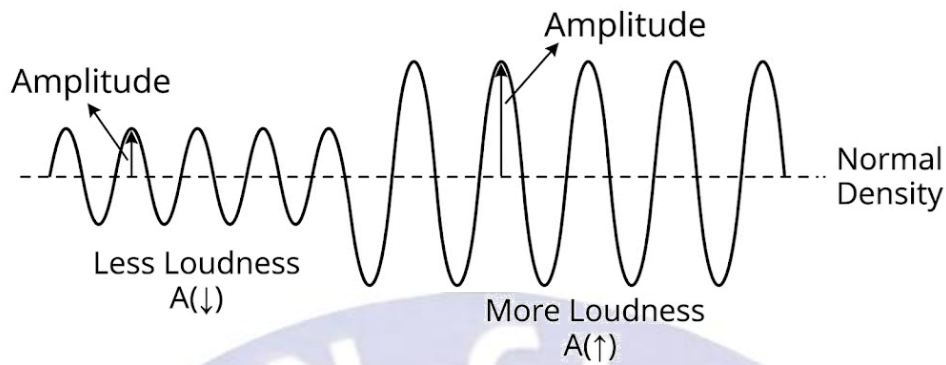
i.e, Loudness $\propto$ Amplitude

Frequency constant



Note:

➤ Loudness of sound decreases with distance only for understand.



- **Amplitude**

- The maximum displacement travelled by the particles of the medium from their mean positions during the wave propagation is called amplitude.
- The unit of amplitude in the term of density is Kg/m^3 and in the term of pressure is N/m^2 , in term of displacement is metre.

Note:

- The amplitude of a sound wave in air is the maximum displacement of a layer of air from its mean position as the sound wave passes through.

3. Tone

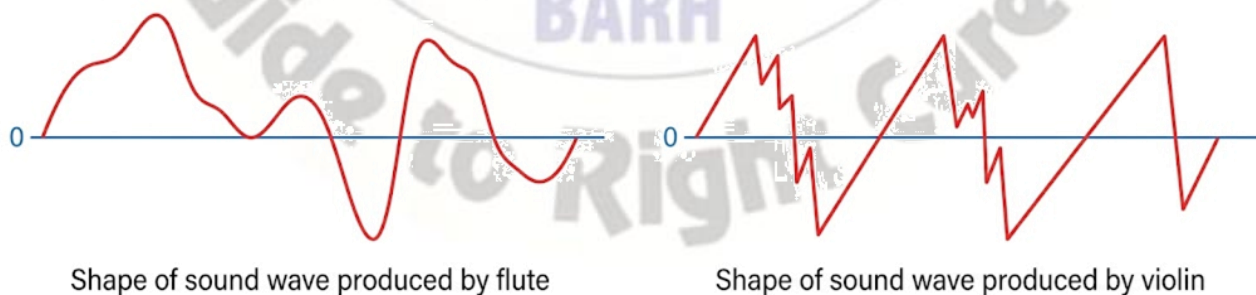
- The sound of a particular frequency is called tone.

Example:-

- ✓ The Indian music Sa, Re, Ga, Ma, Pa, Dha, Ni, Sa represents tones.

- **Timbre**

- The quality of a musical sound which is used for differentiating two sounds when they are in the same frequency.
- Timbre or, quality is that characteristics of musical sound which enables us to distinguish between the sounds of same pitch and loudness produced by different musical instruments (and different singers).
- The timbre or, quality of musical sound depends on the shape of sound wave (waveform) produced by it.



- The quality or, timbre of a musical sound depends on the mixture of frequencies present in it.

Special Note:

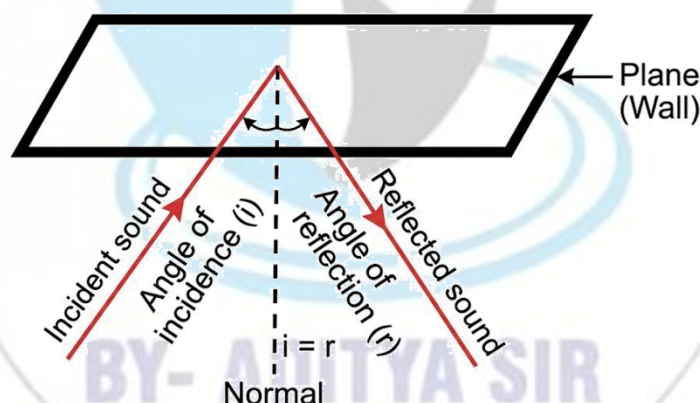
1. Sound travels 15 times faster in iron/steel than in air.
2. Sound travels 5 times faster in water than in air.
3. The speed of light in air is very much greater than the speed of sound in air due to this reason some phenomenon occurred in everyday life.
 - a) In rainy season, the flash of lightning is seen first and the sound of thunder is heard a little later.
 - b) In the cricket, the ball is seen to hit the bat. First and the sound of hitting is heard a little later.
 - c) If a gun is fired from a distance we see the flash of gun first and sound of gun short is heard a little later.
 - d) If the plane flying in the night then its light seen first and sound is heard a little later.

- **Reflection of Sound**

- When sound incidents on a hard surface then it reflects by a certain law, this phenomenon is called reflection of sound and this law is called law of reflection of sound.

OR

The phenomenon of bouncing back of sound when it incidents on a hard surface is called reflection of sound.



There are two laws of reflection of sound

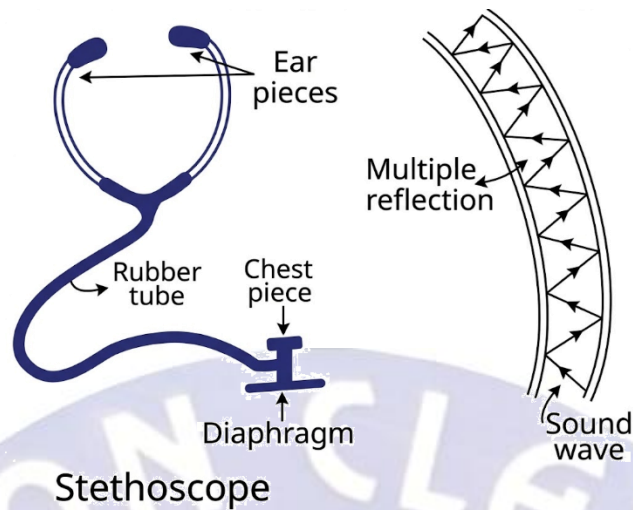
1. Angle of incidence (i) is equal to angle of reflection (r).
2. The incident sound, reflected sound and normal are in the same plane.

- **Applications of reflection of sound**

- Megaphone, Bulbhorn, Stethoscope, soundboard etc.

- **Stethoscope**

- It is a medical instrument used by the doctors for listening to the sounds produced within the human body, mainly in the heart and the lungs.

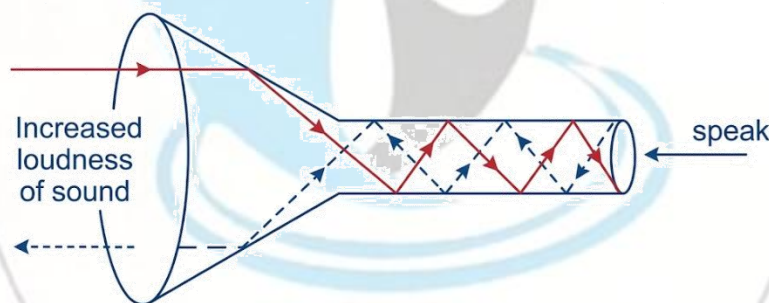


▪ **Working principle of stethoscope**

- A stethoscope works on the principle of multiple reflection of sound.

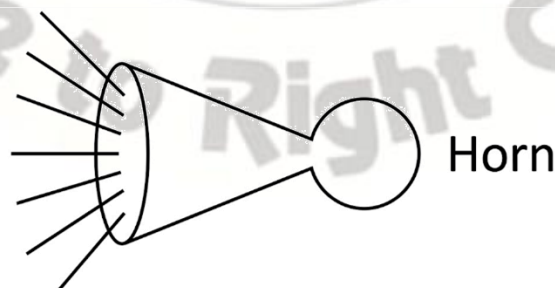
• **Megaphone**

- A megaphone is a large, cone – shaped device for amplifying and directing the voice of a person who speaks into it.
It works on the multiple reflection of sound due to which the loudness of sound increases.
➤ A megaphone is also called “loud – hailer” or “speaking tube”.



• **Horn or Bulb Horn**

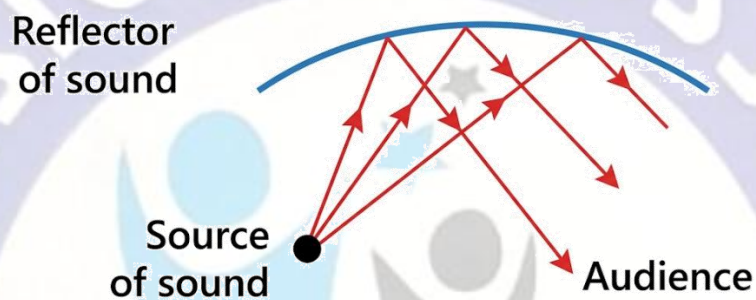
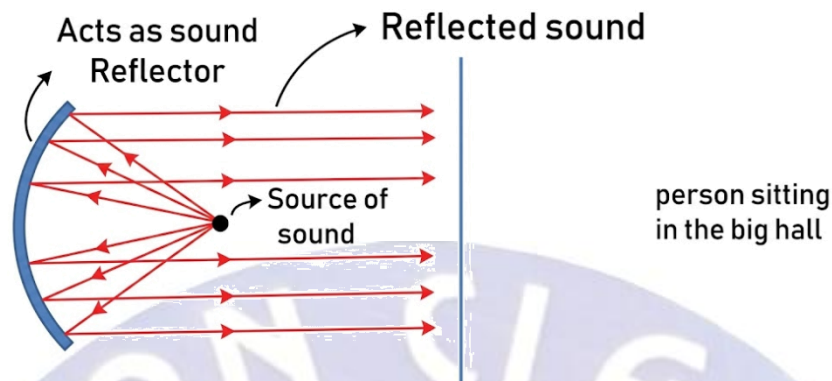
- A bulb horn/horn is a cone – shaped wind instrument which is used for signaling in bicycles, cars, buses, trucks and boats etc.
It also works on the multiple reflection of sound.



• **Sound + board/Sound board**

- The soundboard is a concave board which is placed behind the speaker in large halls or auditoriums so that his speech can heard easily even by the person sitting at a considerable distance.
It also works on multiple reflection of sound.

- The ceiling of concert halls, conference halls and cinema halls are made curved so the sound, after reflection from the ceiling, reaches all the parts of the hall.



- **Reverberation**

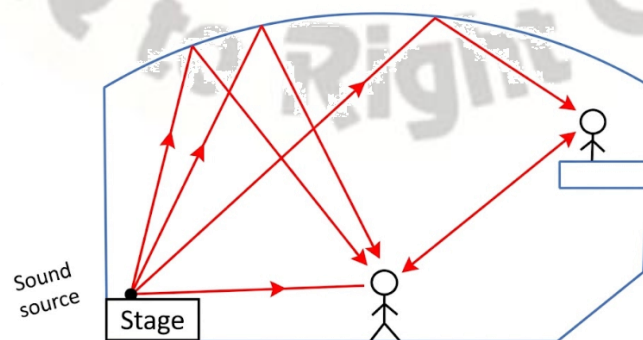
- The persistence of sound due to repeated reflection and its gradual fading away is called reverberation of sound.
- It is created in closed room, hall etc.
- Reverberation can be reduced by using sound proof materials in wall, ceiling, floors etc.

Example of fire proof material:-

- ✓ Wooden panels, matting in floor, foam etc.

Note:

- We only use rough materials for reducing reverberations because soft material absorbs the sound.



Reverberation

Note:

- Furniture are also good sound material to reduces reverberations.

- **“Echo” OR “Goonj” OR “Pratidhavni”**

- The repetition of sound again and again by reflection of sound wave is called an echo.
- The formation of an echo only possible in hills, longs/very big halls etc.
- An echo is formed either in open or closed space.
- An echo of original sound separately only if there is a time interval/time gap of at – least $\left(\frac{1}{10}\right)^{th}$ of a second (0.1 seconds) between then.

For Formation of an echo

The temperature of air = 20°C

Speed of Sound = 344 m/s

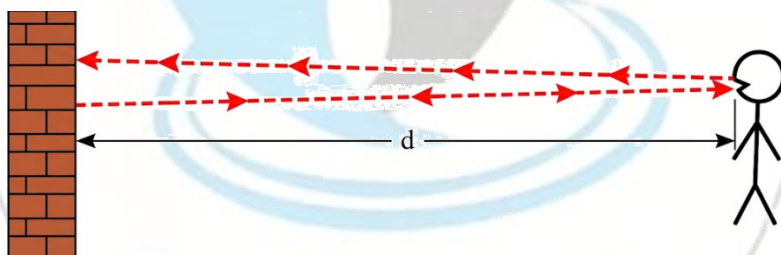
Time taken = 0.1 seconds

$\therefore \text{Distance} = \text{Speed} \times \text{Time}$

$\therefore \text{Required distance for an echo} = 344 \times 0.1 = 34.4 \text{ m}$

$\therefore 2d = 34.4 \text{ m}$

$\therefore d = \frac{34.4}{2} = 17.2 \text{ m}$



Total distance = Man to wall + wall to Man
 $= d + d = 2d$

Note:

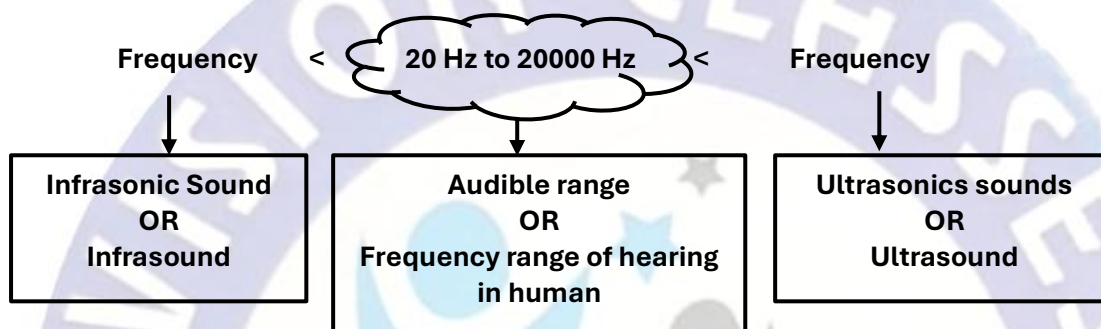
1. So, minimum distance for an echo = 17.2 m
i.e, An echo is formed when $d > 17.2 \text{ m}$ at temperature 20°C
2. An echo is heard sooner on a hot day than a cold day because speed of sound in hot air is more than cold air.
3. The echo distance (17.2 m) will be changed if changed in temperature.

▪ **Differences between Echo and reverberation**

Echo	Reverberation
The distance between source and reflection must be more than 17.2 m	The distance between source and reflection must be less than 17 m
It is the repetition of sound by reflection.	It is the multiple reflection of sound.
The intensity of echo sound is less than that of original sound.	The intensity of reverberated sound is more than its original sound.

• **Audible sound**

- The sound which we are able to hear is called audible sound.
- The audible range of sound frequencies for human ear is from 20 Hz to 20000 Hz.



• **Infrasonic Sound / Infrasound**

- The sound of frequency lower than 20 Hz is called Infrasonic Sound.
- It cannot be heard by humans.
- It can be only heard by some animal like as Rhinos, Hippos, Elephant, Whales, Octopuses, Pigeons, Squid, Cuttlefish, Cod, Guinea fowl etc.

Note:

1. Rhinos, Elephants & Whales can be heard as well as produces infrasonic sound.
2. These animals can hear frequency of earthquakes before it occurs, and they disturb.

• **Ultrasonic Sound / Ultrasound**

- The sound of frequency higher than 20000 Hz is called ultrasonic sound.
- It can't be heard by human.
- It can be heard by dogs, whales, bat, dolphins etc.

Note:

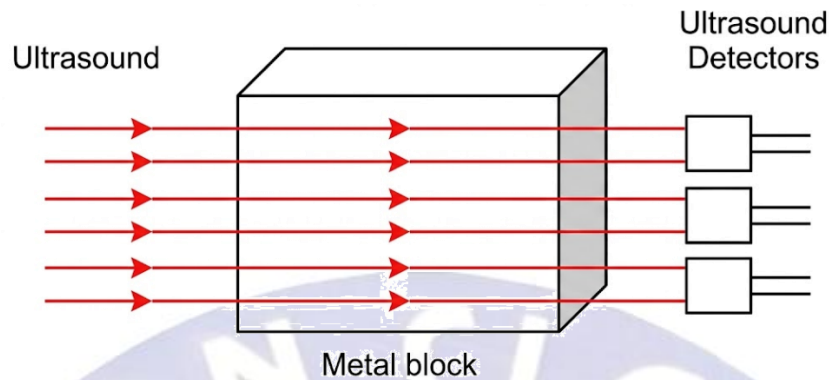
1. Bats can be heard as well as produces ultrasonic sound.
2. The reflection of sound and produces echo also possible in ultrasound.

▪ **Applications of Ultrasound**

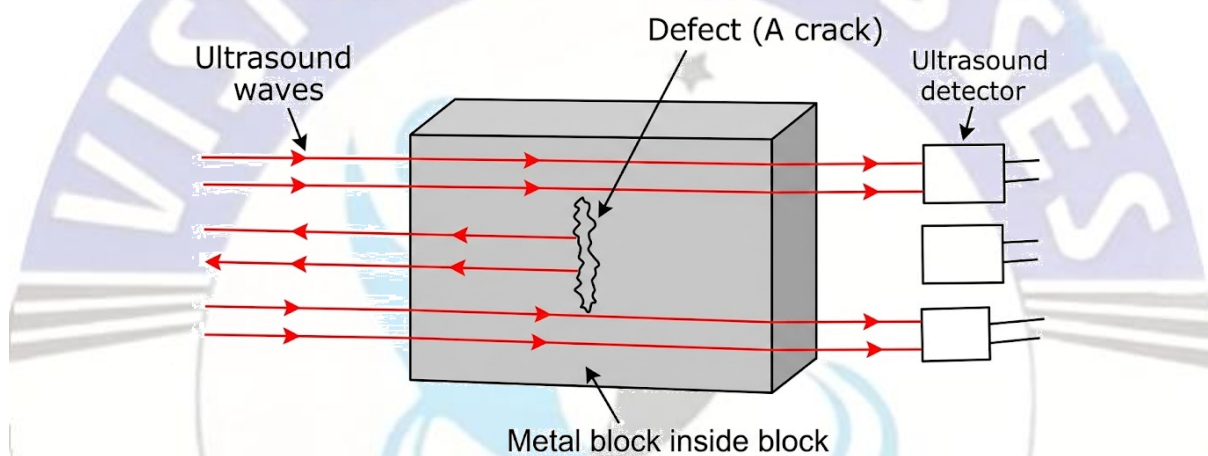
- The ultrasound waves have very high frequency and very short wavelength due to which they can penetrate into "matter" to a large extent i.e, Ultrasounds have penetrating power.

1. Ultrasound is used in industry for detecting cracks in metal blocks without damaging them.

Case 1: When metal blocks defect free OR flawless



Case 2: When metal blocks defect

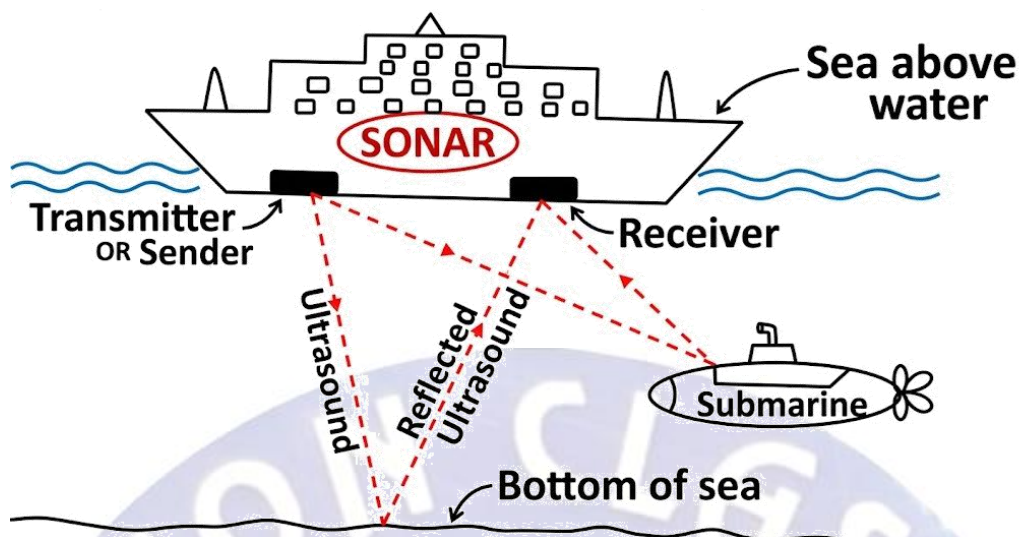


2. Ultrasound is used in industry to clean “hard to reach” parts of objects such as spiral tubes, odd – shaped machines and electronic components etc.
 3. Ultrasound is used to investigate the internal organs of the human body such as liver, gall bladder, pancreas, Kidneys, uterus and heart etc.
- **Ultrasonography**
 - The technique of obtaining pictures of internal echo's of ultrasound pulses is called “Ultrasonography”.
 - **Echocardiography**
 - The use of Ultrasound waves to investigate the action of heart is called echocardiography.
4. Ultrasound scans are used to monitor the development of fetus (unborn baby) inside the mother's uterus.
 5. Ultrasound is used to break Kidney Stones into fine grains (which they get flushed out with urine).
 6. Ultrasound is used in “SONAR” which measures distances which are practically impossible to measure – depth of ocean, distance and direction of underwater objects.

SONAR

Full Form = “SOUND NAVIGATION AND RANGING”

- Sonar is an apparatus / device which is used to find the depth of a sea or to locate the under – water things like shoals of fish, shipwrecks, and enemy submarines.



▪ A sonar has two important parts:-

1. Transmitter (Sending ultrasound).
2. Receiver (Detecting ultrasound echo).

Note:

- The speed of ultrasound in water = the speed of normal sound in water.
 $= 1500 \text{ m/s}$
 $= 1520 \text{ m/s}$
- Time taken by sonar is actual half of time taken by ship to sea level and sea to ship (returning)
 i.e, $t = \frac{\text{Sending time} + \text{receiving time}}{2}$
 Or, $t = \frac{\text{the given time}}{2}$
- ∴ Distance / Length of sea level = Speed of Ultrasound $\times$ Time
 $\Rightarrow \boxed{\text{Distance} = 1500 \times t}$
- This method of finding distance is also called echo ranging.

For example:-

Q. A fishing boat using SONAR detects a school of fish 190 m below it. How much time elapsed between sending the ultrasound signal which detects the fish and receiving the signal's echo? (take speed of ultrasound in water = 1520 m/s).

Ans Since given,

$$d = 190 \text{ m}$$

$$v = 1520 \text{ m/s}$$

$$t =$$

$$\begin{aligned} \text{total distance} &= d + d = 2d \\ &= 190 + 190 \\ &= 380 \text{ m} \end{aligned}$$

$$\therefore t = \frac{d}{v} = \frac{380}{1520} = 0.25 \text{ second}$$

$\therefore$ time taken = 0.25 second

Total distance covered by ultrasound = sending distance + Back distance = $d + d = 2d$

Time taken (go + back) = t

Speed of sound = v

$$\therefore 2d = v \times t \quad \Rightarrow d = \frac{v \times t}{2}$$

Note:

- Ultrasound is used in SONAR because it has penetrating power as well as silent process (Not noisy).

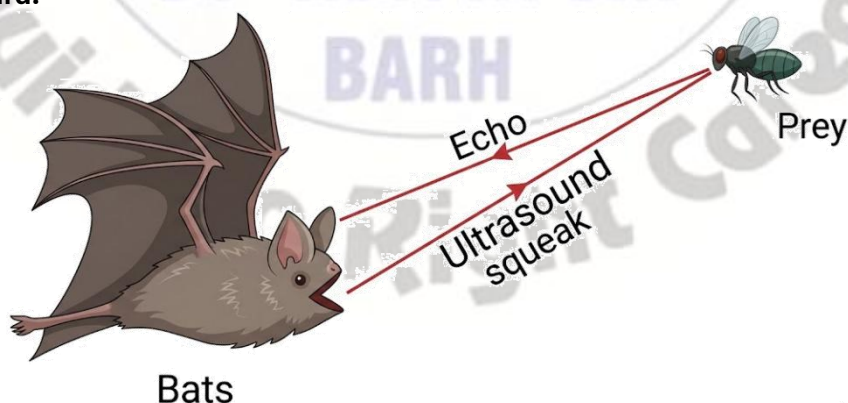
- **Bats Fly and Search Prey Process**

Case 1: Bats Fly Process

- Bats fly in the darkness of night without colliding with other objects by process of echolocation.
In this process bats emit ultrasound squeaks while fly listen to the echo produced by the reflection of their squeaks from the objects in their path. Bats can judge the distance of object in their path and change the path according to dangerous.

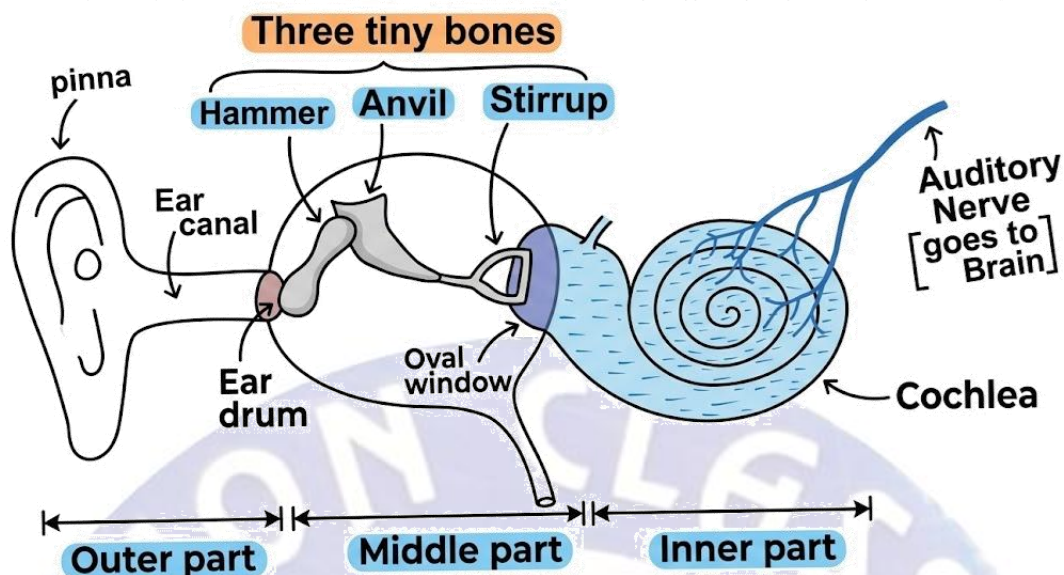
Case 2: Bats Search their Prey Process

- Bats search their prey in night by the process of echolocation.
In this process bats emit ultrasonic squeaks while fly listen to the echo produced by the reflection of their squeaks from the prey (insect) from the time taken by the echo to be heard, bats can judge the distance of the insect to be heard.



- **Human ears**

- Our ears are one of our most important sense organs.
The human ears are divided into three parts; outer ear, middle ear and inner ear.



Structure of Human Ear

- **Outer ear**
 - The outer ear consists of pinna and the ear canal. Pinna is the part which is visible from outside also, visible in the opening of the ear canal, which is about 2 cm long the functional of the outer ear is to guide sound waves to the middle ear.
- **Middle ear**
 - The middle ear is separated from the ear canal by a tightly stretched membrane called the eardrum. Beyond it three interconnected bones called the hammer, anvil and stirrup. The function of the middle is to pick up, amplify and transmit sound waves to the inner ear.
- **Inner ear**
 - The inner ear consists of a liquid filled coiled tube called the cochlea, which is shaped like as snail. This tube is connected to the stirrup. The cochlea has specially sensory cells called hair cells which are so called because of the hairlike structures that stick out of them. The hair cells connected to the auditory nerve, which is connected to the brain.
- **Working of Human ear**
 - When the sound waves fall on ear drum through canal, the eardrum starts vibrating back and forth rapidly because of sound compression & rarefaction in air.

A vibration of the eardrum makes the each of the interconnected bones the hammer, anvil and stirrup to vibrate. These interconnected bones act like levers, which increases the vibration of eardrums more than 20 times.

The vibrating stirrup strikes on the membrane to the oval window and passes its vibration to the liquid in the cochlea due to this, the liquid in the cochlea begins to vibrate. The vibrating liquid of cochlea sets up electrical impulses in the nerve cells present in it. These electrical impulses are carried by auditory nerve to the brain. The brain intercepts these electrical impulses as sound, and we get the sensation of hearing.

Note:

1. The function of three tiny bones in the middle ear is to increase the strength of vibrations coming from the ear – drum before passing them on to the inner ear.
2. We should not put anything (like pin, pencil or pen etc.) inside our ears because they tear our eardrum and make a person deaf.